

Vision-and-Language Navigation: Interpreting visually-grounded navigation instructions in real environments

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Abstract

A robot that can carry out a natural-language instruction has been a dream since before the *Jetsons* cartoon series imagined a life of leisure mediated by a fleet of attentive robot helpers. It is a dream that remains stubbornly distant. However, recent advances in vision and language methods have made incredible progress in closely related areas. This is significant because a robot interpreting a natural-language navigation instruction on the basis of what it sees is carrying out a vision and language process that is similar to Visual Question Answering. Both tasks can be interpreted as visually grounded sequence-to-sequence translation problems, and many of the same methods are applicable. To enable and encourage the application of vision and language methods to the problem of interpreting visually-grounded navigation instructions, we present the *Matterport3D Simulator* – a large-scale reinforcement learning environment based on real imagery [11]. Using this simulator, which can in future support a range of embodied vision and language tasks, we provide the first benchmark dataset for visually-grounded natural language navigation in real buildings – the *Room-to-Room (R2R) dataset*¹.

1. Introduction

The idea that we might be able to give general, verbal instructions to a robot and have at least a reasonable probability that it will carry out the required task is one of the long-held goals of robotics, and artificial intelligence (AI). Despite significant progress, there are a number of major technical challenges that need to be overcome before robots will be able to perform general tasks in the real world. One of the primary requirements will be new techniques for linking natural language to vision and action in *unstructured, previously unseen environments*. It is the navigation version



Instruction: Head upstairs and walk past the piano through an archway directly in front. Turn right when the hallway ends at pictures and table. Wait by the moose antlers hanging on the wall.

Figure 1. Room-to-Room (R2R) navigation task. We focus on executing natural language navigation instructions in previously unseen real-world buildings. The agent’s camera can be rotated freely. Blue discs indicate nearby (discretized) navigation options.

of this challenge that we refer to as Vision-and-Language Navigation (VLN).

Although interpreting natural-language navigation instructions has received significant attention previously [12, 13, 20, 38, 41, 52], it is the recent success of recurrent neural network methods for the joint interpretation of images and natural language that motivates the VLN task, and the associated Room-to-Room (R2R) dataset described below. The dataset particularly has been designed to simplify the application of vision and language methods to what might otherwise seem a distant problem.

Previous approaches to natural language command of robots have often neglected the visual information processing aspect of the problem. Using rendered, rather than real images [7, 27, 62], for example, constrains the set of vis-

¹<https://bringmeaspoon.org>

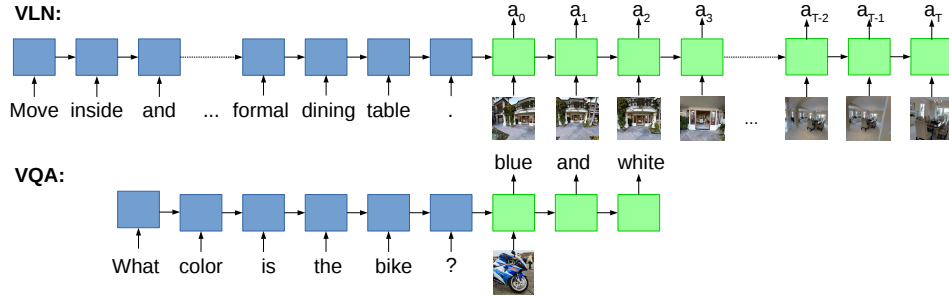


Figure 2. Differences between Vision-and-Language Navigation (VLN) and Visual Question Answering (VQA). Both tasks can be formulated as visually grounded sequence-to-sequence transcoding problems. However, VLN sequences are much longer and, uniquely among vision and language benchmark tasks using real images, the model outputs actions $\langle a_0, a_1, \dots, a_T \rangle$ that manipulate the camera viewpoint.

ible objects to the set of hand-crafted models available to the renderer. This turns the robot’s challenging open-set problem of relating real language to real imagery into a far simpler closed-set classification problem. The natural extension of this process is that adopted in works where the images are replaced by a set of labels [13, 52]. Limiting the variation in the imagery inevitably limits the variation in the navigation instructions also. What distinguishes the VLN challenge is that the agent is required to interpret a previously *unseen* natural-language navigation command in light of images generated by a previously *unseen* real environment. The task thus more closely models the distinctly open-set nature of the underlying problem.

To enable the reproducible evaluation of VLN methods, we present the Matterport3D Simulator. The simulator is a large-scale interactive reinforcement learning (RL) environment constructed from the Matterport3D dataset [11] which contains 10,800 densely-sampled panoramic RGB-D images of 90 real-world building-scale indoor environments. Compared to synthetic RL environments [7, 27, 62], the use of real-world image data preserves visual and linguistic richness, maximizing the potential for trained agents to be transferred to real-world applications.

Based on the Matterport3D environments, we collect the Room-to-Room (R2R) dataset containing 21,567 open-vocabulary, crowd-sourced navigation instructions with an average length of 29 words. Each instruction describes a trajectory traversing typically multiple rooms. As illustrated in Figure 1, the associated task requires an agent to follow natural-language instructions to navigate to a goal location in a previously unseen building. We investigate the difficulty of this task, and particularly the difficulty of operating in unseen environments, using several baselines and a sequence-to-sequence model based on methods successfully applied to other vision and language tasks [4, 14, 19].

In summary, our main contributions are:

1. We introduce the Matterport3D Simulator, a software framework for visual reinforcement learning using the

Matterport3D panoramic RGB-D dataset [11];

2. We present Room-to-Room (R2R), the first benchmark dataset for Vision-and-Language Navigation in real, previously unseen, building-scale 3D environments;
3. We apply sequence-to-sequence neural networks to the R2R dataset, establishing several baselines.

The simulator, R2R dataset and baseline models are available through the project website at <https://bringmeaspoon.org>.

2. Related Work

Navigation and language Natural language command of robots in unstructured environments has been a research goal for several decades [57]. However, many existing approaches abstract away the problem of visual perception to some degree. This is typically achieved either by assuming that the set of all navigation goals, or objects to be acted upon, has been enumerated, and that each will be identified by label [13, 52], or by operating in visually restricted environments requiring limited perception [12, 20, 24, 29, 35, 38, 55]. Our work contributes for the first time a navigation benchmark dataset that is both linguistically and visually rich, moving closer to real scenarios while still enabling reproducible evaluations.

Vision and language The development of new benchmark datasets for image captioning [14], visual question answering (VQA) [4, 19] and visual dialog [17] has spurred considerable progress in vision and language understanding, enabling models to be trained end-to-end on raw pixel data from large datasets of natural images. However, although many tasks combining visual and linguistic reasoning have been motivated by their potential robotic applications [4, 17, 26, 36, 51], none of these tasks allow an agent to move or control the camera. As illustrated in Figure 2, our proposed R2R benchmark addresses this limitation, which also motivates several concurrent works on embodied question answering [16, 18].

Navigation based simulators Our simulator is related to existing 3D RL environments based on game engines, such as ViZDoom [27], DeepMind Lab [7] and AI2-THOR [30], as well as a number of newer environments developed concurrently including HoME [10], House3D [58], MINOS [47], CHALET [59] and Gibson Env [61]. The main advantage of our framework over synthetic environments [30, 10, 58, 59] is that all pixel observations come from natural images of real scenes, ensuring that almost every coffee mug, pot-plant and wallpaper texture is unique. This visual diversity and richness is hard to replicate using a limited set of 3D assets and textures. Compared to MINOS [47], which is also based on Matterport data [11], we render from panoramic images rather than textured meshes. Since the meshes have missing geometry – particularly for windows and mirrors – our approach improves visual realism but limits navigation to discrete locations (refer Section 3.2 for details). Our approach is similar to the (much smaller) Active Vision Dataset [2].

RL in navigation A number of recent papers use reinforcement learning (RL) to train navigational agents [31, 50, 53, 62, 21], although these works do not address language instruction. The use of RL for language-based navigation has been studied in [12] and [41], however, the settings are visually and linguistically less complex. For example, Chaplot *et al.* [12] develop an RL model to execute template-based instructions in Doom environments [27]. Misra *et al.* [41] study complex language instructions in a fully-observable blocks world. By releasing our simulator and dataset, we hope to encourage further research in more realistic partially-observable settings.

3. Matterport3D Simulator

In this section we introduce the Matterport3D Simulator, a new large-scale visual reinforcement learning (RL) simulation environment for the research and development of intelligent agents based on the Matterport3D dataset [11]. The Room-to-Room (R2R) navigation dataset is discussed in Section 4.

3.1. Matterport3D Dataset

Most RGB-D datasets are derived from video sequences; e.g. NYUv2 [42], SUN RGB-D [48] and ScanNet [15]. These datasets typically offer only one or two paths through a scene, making them inadequate for simulating robot motion. In contrast to these datasets, the recently released Matterport3D dataset [11] contains a comprehensive set of panoramic views. To the best of our knowledge it is also the largest currently available RGB-D research dataset.

In detail, the Matterport3D dataset consists of 10,800 panoramic views constructed from 194,400 RGB-D images of 90 building-scale scenes. On average, panoramic view-

points are distributed throughout the entire walkable floor plan of each scene at an average separation of 2.25m. Each panoramic view is comprised of 18 RGB-D images captured from a single 3D position at the approximate height of a standing person. Each image is annotated with an accurate 6 DoF camera pose, and collectively the images capture the entire sphere except the poles. The dataset also includes globally-aligned, textured 3D meshes annotated with class and instance segmentations of regions (rooms) and objects.

In terms of visual diversity, the selected Matterport scenes encompass a range of buildings including houses, apartments, hotels, offices and churches of varying size and complexity. These buildings contain enormous visual diversity, posing real challenges to computer vision. Many of the scenes in the dataset can be viewed in the Matterport 3D spaces gallery².

3.2. Simulator

3.2.1 Observations

To construct the simulator, we allow an embodied agent to virtually ‘move’ throughout a scene by adopting poses coinciding with panoramic viewpoints. Agent poses are defined in terms of 3D position $v \in V$, heading $\psi \in [0, 2\pi)$, and camera elevation $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where V is the set of 3D points associated with panoramic viewpoints in the scene. At each step t , the simulator outputs an RGB image observation o_t corresponding to the agent’s first person camera view. Images are generated from perspective projections of precomputed cube-mapped images at each viewpoint. Future extensions to the simulator will also support depth image observations (RGB-D), and additional instrumentation in the form of rendered object class and object instance segmentations (based on the underlying Matterport3D mesh annotations).

3.2.2 Action Space

The main challenge in implementing the simulator is determining the state-dependent action space. Naturally, we wish to prevent agents from teleporting through walls and floors, or traversing other non-navigable regions of space. Therefore, at each step t the simulator also outputs a set of next step reachable viewpoints $W_{t+1} \subseteq V$. Agents interact with the simulator by selecting a new viewpoint $v_{t+1} \in W_{t+1}$, and nominating camera heading ($\Delta\psi_{t+1}$) and elevation ($\Delta\theta_{t+1}$) adjustments. Actions are deterministic.

To determine W_{t+1} , for each scene the simulator includes a weighted, undirected graph over panoramic viewpoints, $G = \langle V, E \rangle$, such that the presence of an edge signifies a robot-navigable transition between two viewpoints,

²<https://matterport.com/gallery/>